51.504 Machine Learning (2025)
Homework 3

Due 9 Dec 2025, Tuesday, 11.59pm

## Question 1

(a) [10 points] What is the output when we convolve the left image with the $3 \times 3$ filter on the right (assuming stride = 1 and no padding)?

$$
\begin{array}{|c|c|c|c|c|}
\hline
1 & 1 & 2 & 0 & 3 \\ \hline
2 & 4 & 2 & 1 & 1 \\ \hline
1 & 1 & 5 & 3 & 0 \\ \hline
1 & 2 & 6 & 0 & 1 \\ \hline
0 & 1 & 2 & 0 & 3 \\ \hline
\end{array}
\times
\begin{array}{|c|c|c|}
\hline
1 & 0 & -1 \\ \hline
0 & 1 & 0 \\ \hline
-1 & 0 & 1 \\ \hline
\end{array}
=
\begin{array}{|c|c|c|}
\hline
\phantom{0} & \phantom{0} & \phantom{0} \\ \hline
\phantom{0} & \phantom{0} & \phantom{0} \\ \hline
\phantom{0} & \phantom{0} & \phantom{0} \\ \hline
\end{array}
$$

(b) [10 points] What is the output if we instead perform convolution using padding = 1, stride = 2, and then apply ReLU?

## Question 2

Let $X = (X_1, X_2, X_3, X_4)$ be the random variable with $0 \leq X_i \leq 4$ for $i = 1, 2, 3, 4$, and density function

$$f_X(x) = \frac{1}{Z_1} \exp\{x_1 x_2 x_3 x_4\}, \quad Z_1 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{x_1 x_2 x_3 x_4\}.$$

Let $Y = (Y_1, Y_2, Y_3, Y_4)$ be the random variable with $0 \leq Y_i \leq 4$ for $i = 1, 2, 3, 4$, and density function

$$f_Y(y) = \frac{1}{Z_2} \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}, \quad Z_2 = \int_0^4 \int_0^4 \int_0^4 \int_0^4 \exp\{y_1 y_2 y_3 + y_1 y_2 + y_4\}.$$

Let $G$ be the graph in Figure 1.

(a) [5 points] Show that $G$ is a Markov random field for $Y$ but not $X$.

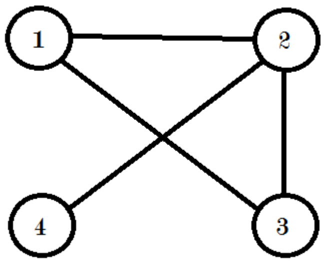


Figure 1: Graph G

(b) [5 points] Write down a conditional independence relationship for $Y$ that can be read off the graph $G$ due to the global Markov property. (Note that $Y_i$ is the random variable associated with node $i$.)

(c) [5 points] Write down the conditional independence relationship for $Y$ that follows from the local Markov property for node 4.

(d) [5 points] Draw the appropriate arrows on $G$ for the Bayesian network associated with the following set of structural equations:

$$
\begin{aligned}
Y_1 &= -2Y_3 + \epsilon_1, \\
Y_2 &= 4Y_1 + Y_3 + \epsilon_2, \\
Y_3 &= \epsilon_3, \\
Y_4 &= -3Y_2 + \epsilon_4, \\
\epsilon_i &\sim \mathcal{N}(0, 1).
\end{aligned}
$$

## Question 3

Consider a mixture of two Gaussians, $\mathcal{N}(\mu_1, \sigma_1^2)$ with density function $f_1(x)$ and proportion $\pi_1$, and $\mathcal{N}(\mu_2, \sigma_2^2)$ with density function $f_2(x)$ and proportion $\pi_2$. The EM algorithm iteratively updates the parameters with the E-step,

$$\gamma(z_k^{(i)}) = \frac{\pi_k C_k \exp\{-(x^{(i)} - \mu_k)^2/2\sigma_k^2\}}{\sum_{j=1}^{2} \pi_j C_j \exp\{-(x^{(i)} - \mu_j)^2/2\sigma_j^2\}},$$

and the M-step,

$$\pi_k = \frac{1}{2}\sum_{i=1}^{2}\gamma(z_k^{(i)}),$$

$$\mu_k = \frac{\sum_{i=1}^{2} x^{(i)}\gamma(z_k^{(i)})}{\sum_{i=1}^{2}\gamma(z_k^{(i)})},$$

$$\sigma_k^2 = \frac{\sum_{i=1}^{2}(x^{(i)}-\mu_k)^2\gamma(z_k^{(i)})}{\sum_{i=1}^{2}\gamma(z_k^{(i)})}.$$

Suppose we perform EM on the sample data set $\{1, 1.5\}$. Let our initial parameter estimates be $\mu_1 = 0, \mu_2 = 3, \sigma_1 = 1, \sigma_2 = 2$. Also let our initial mixture parameters be $\pi_1 = 1/3$. We then have $f_1(1) = 0.241$, $f_1(1.5) = 0.13$, $f_2(1) = 0.12$ and $f_2(1.5) = 0.15$.

(a) [10 points] Calculate the probabilities that each data point belongs to each Gaussian given the initial parameters.

(b) [10 points] Compute the next iteration of $\pi_1, \pi_2, \mu_1, \mu_2$.

(c) [10 points] Compute the next iteration of $\sigma_1^2$ and $\sigma_2^2$

## Question 4

Consider the irregular neural network N in Figure 2. Let i0, i1 and i2 be the input nodes and o0, o1, o2 be the output nodes. Let $x_0$, $x_1$ and $x_2$ be the input variables of i0, i1 and i2 respectively. Let the activation function at nodes h0, h1, o0, o1 and o2 be $\sigma(x)$.

(a) [10 points] Write down the output of o2 in terms of x, w and $\sigma$.

(b) [5 points] Which nodes' error signals $\delta$ contains the weight w1?

For the remainder part of the question, we treat network N as a graph. Please also list down all possible paths and show which paths are open or closed.

(c) [5 points] Is h0 d-separated from h1?

(d) [5 points] Is i1 d-separated from o1?

(e) [5 points] Is i1 d-separated from o1 given w1, w2 and o2?

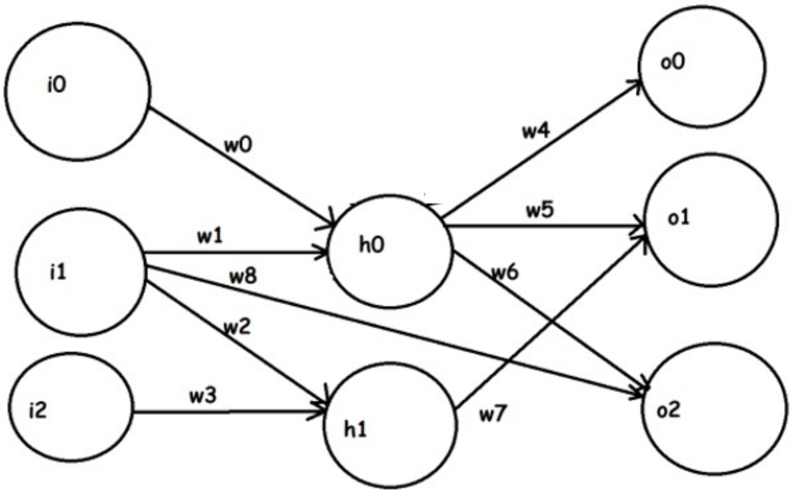


Figure 2: Network N

# Solutions:

## Question 1

(a)

| | | |
|---|---|---|
| 7 | 5 | -5 |
| 6 | 6 | -1 |
| 0 | 3 | 6 |

(b)

| | | |
|---|---|---|
| 5 | 0 | 2 |
| 0 | 6 | 1 |
| 0 | 4 | 3 |

## Question 2

(a) $Y$ factorizes according to the cliques of $G$, which are $\{1, 2, 3\}$ and $\{2, 4\}$, as

$$f_Y(y) = \frac{1}{Z_2} \exp\{y_1 y_2 y_3 + y_1 y_2\} \cdot \exp\{y_4\}.$$

$X$ does not factorize according to the cliques of $G$, and $X_4 \not\perp X_1, X_3 \,|\, X_2$, so it fails the global Markov property.

(b) $Y_4 \perp Y_1, Y_3 \mid Y_2$ (there are other conditional independence relationships as well).

(c) $Y_4 \perp Y_1 \mid Y_2$. Can also be $Y_4 \perp Y_3 \mid Y_2$, or $Y_4 \perp Y_1, Y_3 \mid Y_2$

(d) Refer to Figure 2.

# Question 3

(a) For data point 1, we have $\pi_1 f_1(1)+\pi_2 f_2(1) = 1/3 \cdot 0.241+2/3 \cdot 0.12 = 0.160$. Therefore, we have $\gamma(z_1^{(1)}) = \pi_1 f_1(1)/0.160 = 0.5$ and $\gamma(z_2^{(1)}) = \pi_2 f_2(1)/0.160 = 0.5$. For data point 1.5, we have $\pi_1 f_1(1.5) + \pi_2 f_2(1.5) = 1/3 \cdot 0.13 + 2/3 \cdot 0.15 = 0.143$. Therefore, we have $\gamma(z_1^{(1.5)}) = \pi_1 f_1(1.5)/0.143 = 0.3$ and $\gamma(z_2^{(1.5)}) = \pi_2 f_2(1.5)/0.143 = 0.7$

(b) $\pi_1 = 1/2 \cdot (\gamma(z_1^{(1)}) + \gamma(z_1^{(1.5)})) = 0.4$. $\pi_2 = 1/2 \cdot (\gamma(z_2^{(1)}) + \gamma(z_2^{(1.5)})) = 0.6$. $\mu_1 = (1 \cdot 0.5 + 1.5 \cdot 0.3)/0.8 = 1.19$. $\mu_2 = (1 \cdot 0.5 + 1.5 \cdot 0.7)/1.2 = 1.29$.

(c) $\sigma_1^2 = [0.5 \cdot (1 - 1.19)^2 + 0.3 \cdot (1.15 - 1.19)^2]/0.8 = 0.0586$ and $\sigma_2^2 = [0.5 \cdot (1 - 1.29)^2 + 0.7 \cdot (1.5 - 1.29)^2]/1.2 = 0.0608$.

# Question 4

(a) Output of o2 $= \sigma\{w6[\sigma(w0x_0 + w1x_1)] + w8x_1\}$.

(b) h0, o1 and o2.

(c) No, there is an open path going from h0 to i1 to h1. The other paths are $\{h0, o1, h1\}$ which is closed because of the collider at o1 and $\{h0, o2, i1, h1\}$ which is also closed due to the collider at o2.

(d) No, there is an open path $\{i1, h0, o1\}$. The two other paths are $\{i1, h1, o1\}$ which is open and $\{i1, o2, h0, o1\}$ which is closed due to the collider at o2.

(e) No, since $\{i1, o2, h0, o1\}$ is now open due to the collider o2 being conditioned. The other paths stated in part d are closed.

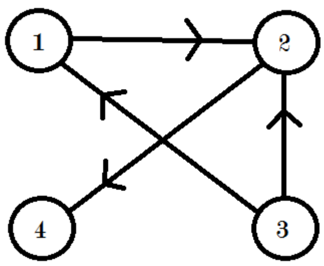


Figure 3: Directed Graph G